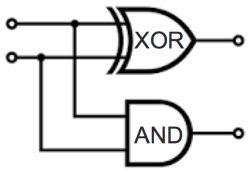
# Half adder circuit

**Fixed-width addition**: adding one bit at time, using the usual column-by-column and carry arithmetic, and dropping the carry from the leftmost column so the result is the same width as the summands. In many cases, this gives representation of the correct value for the sum when we interpret the summands in fixed-width binary or in 2s complement.

**Addition**: For single column:

| Input | | Output | |
|---|---|---|---|
| $x_0$ | $y_0$ | $c_0$ | $s_0$ |
| 1 | 1 | | |
| 1 | 0 | | |
| 0 | 1 | | |
| 0 | 0 | | |




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# Two bit adder circuit

Draw a logic circuit that implements binary addition of two numbers that are each represented in fixed-width binary:

- Inputs $x_0, y_0, x_1, y_1$ represent $(x_1x_0)_{2,2}$ and $(y_1y_0)_{2,2}$
- Outputs $z_0, z_1, z_2$ represent $(z_2z_1z_0)_{2,3} = (x_1x_0)_{2,2} + (y_1y_0)_{2,2}$ (may require up to width 3)

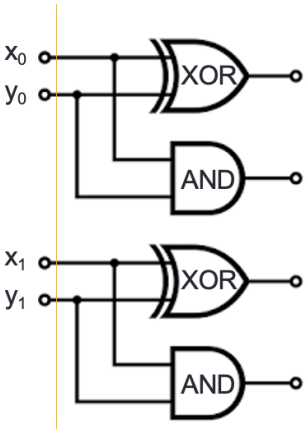
*First approach*: half-adder for each column, then combine carry from right column with sum of left column

Write expressions for the circuit output values in terms of input values:

$z_0 =$ _____

$z_1 =$ _____

$z_2 =$ _____




*There are other approaches, for example*: for middle column, first add carry from right column to $x_1$, then add result to $y_1$

Extra practice: how does this circuit change if we're representing fixed-width addition?

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# Fixed width definition

**Definition** For $b$ an integer greater than 1, $w$ a positive integer, and $n$ a nonnegative integer _____, the **base** $b$ **fixed-width** $w$ **expansion of** $n$ is

$$(a_{w-1} \cdots a_1 a_0)_{b,w}$$

where $a_0, a_1, \ldots, a_{w-1}$ are nonnegative integers less than $b$ and

$$n = \sum_{i=0}^{w-1} a_i b^i$$

# Representing zero

**Representing** 0: Which of the representations so far allow us to represent 0?

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